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Fall 2024 Applied PDE's 440/540 Schreyer

TEST I

1. Please PRINT your name at the top of this page.
2. Check to see that you have 7 pages and 6 problems. There are 2 blank pages for work should you need it.
3. If you do your final work on another page, please indicate clearly where you do your work.
4. One side of an 8.5" by 11" sheet of paper is allowed for notes (no examples). Please turn this page in with the exam.
5. No technology.
6. If you run out of time, explain what you would do if you did have time.
7. Show what you know.

Page 2	18	(18)
Page 3	8	(8)
Pages 4,5	20	(20)
Page 6,7	20	(20)

Total 66 (66) :)

open & connected

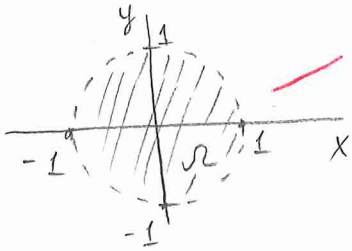
6

1. (6 points) Give an example of a set in $\mathbb{R}^2$ that is a domain using bracket {} notation. Sketch your domain.

$\Omega = \{ (x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1 \}$

Note that ① Ω is an open set &

② Ω is connected

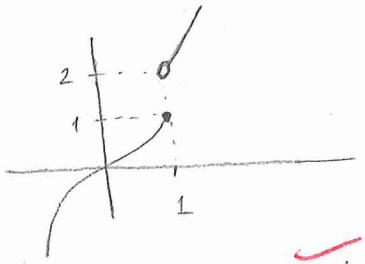


6

2. (6 pts) Let $\mathbb{R}$ be the set of all points on the real line. You may give an algebraic expression of your function or provide a graph (not necessary to provide both).

(a) Give an example of a function that is not in $C^0(\mathbb{R})$.

$$f(x) = \begin{cases} x^3, & x \le 1 \\ 2x, & x > 1 \end{cases}$$



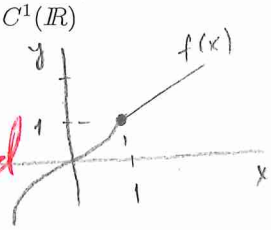
Note the given (required) domain is $\mathbb{R}$, but the function $f(x) \notin C^0(\mathbb{R})$, discon-s at $x=1$.

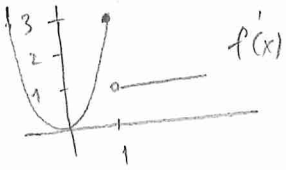
(b) Give an example of a function that is in $C^0(\mathbb{R})$ but is not in $C^1(\mathbb{R})$

$$f(x) = \begin{cases} x^3, & x \le 1 \\ x, & x > 1 \end{cases} \in C^0(\mathbb{R})$$

$$f'(x) = \begin{cases} 3x^2, & x \le 1 \\ 1, & x > 1 \end{cases} \notin C^0(\mathbb{R})$$

good





(c) Give an example of a function that is in $C^\infty(\mathbb{R})$

$f(x) = x^3, \quad x \in \mathbb{R}.$

good

6

3. (6 pts) Classify the following first-order PDE's as either linear, almost linear, quasi-linear, or nonlinear. Circle one.

(a) $\overset{P(x,y)}{x}u_x - \overset{Q(x,y)}{xe^y}u_y = \overset{R(x,y,u)}{u^2}$ $\quad u = u(x,y), \; u^2$

A) linear B) almost linear C) quasi-linear D) nonlinear

(b) $\overset{P(x,y)}{x}u_x - \overset{Q(x,y,u)}{xe^u}u_y = \overset{R(x,y,u)}{u^2}$

A) linear B) almost linear C) quasi-linear D) nonlinear

(c) $\overset{P(x,y)}{x}u_x - \overset{Q(x,y)}{xe^y}u_y = \overset{R(x,y,u)}{u + y^3}$ $\Rightarrow$ $\overset{P(x,y)}{(x)}u_x - \overset{Q(x,y)}{(xe^y)}u_y + \overset{R(x,y)}{(-1)}\cdot u = \overset{d(x,y)}{y^3}$

A) linear B) almost linear C) quasi-linear D) nonlinear

8

4. (8 pts) Find $f'(t)$ using Leibniz rule (no points for not using Leibniz rule):

(1) $$f(t) = \int_0^{t^2} xe^{3t}\,dx$$

Solution:

Recall the Leibniz rule:

(2) $$\frac{d}{dt}\int_{g(t)}^{h(t)} f(x,t)\,dx = \int_{g(t)}^{h(t)} \frac{\partial}{\partial t} f(x,t)\,dx + f(h(t),t)\cdot h'(t) - f(g(t),t)\cdot g'(t).$$

Applying this to our function $f(t)$ in (1):

$$f'(t) = \frac{d}{dt}f(t) = \frac{d}{dt}\int_0^{t^2} x\cdot e^{3t}\,dx = \int_0^{t^2}\frac{\partial}{\partial t}\left[x\cdot e^{3t}\right]dx +$$

$$+\left[t^2\cdot e^{3t}\right]\cdot(t^2)' - \underbrace{\left[0\cdot e^{3t}\right]\cdot(0)'}_{0} = \int_0^{t^2} x\cdot\left[3e^{3t}\right]dx + \left[t^2\cdot e^{3t}\right]\cdot 2t - 0$$

$$= \int_0^{t^2} 3xe^{3t}\,dx + 2t^3e^{3t}.$$

It's not given to simplify untill end unlike in HW

So it's ok to stop here, but let me continue in case!

$$= \left[3e^{3t}\frac{x^2}{2}\right]_0^{t^2} + 2t^3e^{3t} = \frac{3e^{3t}}{2}\cdot\left[t^4 - 0^2\right] + 2t^3e^{3t} = \boxed{\frac{3t^4}{2}e^{3t} + 2t^3e^{3t}}$$

Checking (without Leibniz rule):

$$f(t) = \int_0^{t^2} xe^{3t}\,dx = \left[e^{3t}\cdot\frac{x^2}{2}\right]_0^{t^2} = \frac{e^{3t}}{2}\left[(t^2)^2 - 0^2\right] = \frac{t^4\cdot e^{3t}}{2}.\ \text{Then}$$

$$f'(t) = \frac{d}{dt}\left[t^4\cdot\frac{e^{3t}}{2}\right] = \frac{1}{2}\left[(t^4)'\cdot e^{3t} + t^4(e^{3t})'\right] = \frac{4t^3}{2}e^{3t} + \frac{3t^4}{2}e^{3t} =$$

good!

$$= \boxed{2t^3e^{3t} + \frac{3t^4}{2}e^{3t}}$$

same!

20

5. (20 pts) Consider the following first-order PDE with an initial curve:

$$zz_x + z_y = z, \qquad z(x,0) = x.$$

(a) (5 pts) Verify that the solution exists and is unique. Where are we guaranteed that the solution exists?

(b) (12 pts) Solve explicitly for z using the Method of Characteristics (zero points for not using the Method of Characteristics). **Please do NOT draw the characteristics! (no points).**

(c) (3 pts) Check that your solution works.

Note: There is a blank page following this page so that you can continue your work on the following page.

Solution: given

(1) $z z_x + z_y = z$ (Quasi-linear 1st order PDE)

(2) $z(x,0) = x$ (Note no domain defined yet!)

(a) For method of characteristics which is asked in part (b), let me start the steps, along which we answer the question asked in part (a):

① Parametrize the Initial Curve (IC) in (2):

(3) $\langle x, y, z \rangle = \langle x, 0, x \rangle = \langle \tau, 0, \tau \rangle$ (No domain)

IC (2)

②. Identify P, Q, R, f from (1):

(4) $\boxed{P(x,y,z) = z, \quad Q(x,y,z) = 1, \quad R(x,y,z) = z, \quad f(x) = x}$

are all from $C^1(\Omega)$ [?], where unknown domain Ω can be $\mathbb{R}^3$ for now, since these are polynomials. I might be redefining Ω more in detail in the next step.

③ Check the existance & uniqueness theorem requirements using P, Q, R and param-d IC defined in steps ① & ②:

(i) $P(x,y,z) = z$, $Q(x,y,z) = 1$, $R(x,y,z) = z$, $f(x) = x$ are all in $C^1(\Omega)$?, where $\Omega = \mathbb{R}^3$ for now.

(ii) Determinant condition:

We should check this for $\langle x_0(\tau), y_0(\tau), z_0(\tau) \rangle = \langle \overset{x_0}{\tau}, \overset{y_0}{0}, \overset{z_0}{\tau} \rangle$

(5) $$\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0'(\tau), & y_0'(\tau) \end{vmatrix} \neq 0$$

(6) $$\begin{vmatrix} z|_{\langle \tau, 0, \tau \rangle} & 1|_{\langle \tau, 0, \tau \rangle} \\ (\tau)' & 0' \end{vmatrix} = \begin{vmatrix} \tau & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0$$

The determinant $\neq 0$ regardless of choice of τ, so τ (recall the param-n, $\tau = x$) $\in \mathbb{R} \Rightarrow x \in \mathbb{R} \Rightarrow z = x \in \mathbb{R}$, and of course $y \in \mathbb{R}$, i.e. $(x,y,z) \in \mathbb{R}^3$ for determinant condition. In part (i), our temporary domain $\Omega = \mathbb{R}^3$, and here determinant condition is also OK on this $\Omega = \mathbb{R}^3$ domain, so

(7). $\boxed{\Omega = \mathbb{R}^3}$

In summary, in all $\Omega = \mathbb{R}^3$ we're guaranteed the existance & uniqueness of the solution by the theorem.

(b) Solving the (1) + (2) using method of charac-s:
Since, the steps ① – ③ were already given in part (a),

Thank you

(I'm going to add pages from here indexing as $\boxed{5_2}$, $\boxed{5_3}$, and so on....

6. (20 pts) Consider the conservation law equation

$$\overset{a(z)}{-z^2} z_x + z_y = 0, \qquad z(x,0) = \overset{f(x)}{\sin(x)}, \quad y > 0.$$

(1) (2)

It is NOT necessary to show that the solution exists and is unique. This is already in a form that we have checked - we already know the solution exists and is unique

4 (a) (4 pts) Find an *implicit* form of the solution (do not need to solve for the dependent variable).

12 (b) (12 pts) Graph four characteristics corresponding to $\tau = -1,\ 0,\ 1,\ 2$. How does the solution behave along each characteristic?

4 (c) (4 pts) Do shocks occur based on your graph of characteristics?

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Sol-n (a) Recall that the implicit sol-n is given by

$$z = f(x - a(z)\cdot y) = f(x - (-z^2)\cdot y) =$$

$$= f(x + z^2 y) = \sin(x + z^2 y). \Rightarrow$$

(3) $$\boxed{z = \sin(x + z^2 y)}$$

(b) Characteristics are

(4) $$x - x_0 = a(z_0)\, y \Rightarrow x_0 = \tau$$

$$y = \frac{x - x_0}{a(\underbrace{z(x_0)}_{f(x_0)})} = \frac{x-\tau}{a(f(\tau))} = \frac{x-\tau}{a(\sin(\tau))} \qquad \boxed{a(z) = -z^2}$$

$$= \frac{x-\tau}{-\sin^2(\tau)} = -\frac{x}{\sin^2\tau} + \frac{\tau}{\sin^2(\tau)} = y$$

τ	-1	0	1
$y(x)$	$y(x) = -\frac{1}{\sin^2(-1)}x - \frac{1}{\sin^2(-1)}$	$x = 0$	
	(A)	(B)	(C)

please see the 6_2 & 6_3 pages at the end for the rest of answers! Thank you

here let me continue from step ④:

④ Finding the characteristic curves

$$\frac{dx}{ds} = P \qquad \frac{dy}{ds} = Q \qquad \frac{dz}{ds} = R \quad \Rightarrow$$

$$x_0(s=0) = x_0(\tau) \qquad y(s=0) = y_0(\tau) \qquad z(s=0) = z_0(\tau)$$

$$\begin{cases} \frac{dx}{ds} = z \\ x_0(s=0) = \tau \end{cases} \quad (8) \qquad \begin{cases} \frac{dy}{ds} = 1 \\ y_0(s=0) = 0 \end{cases} \quad (9) \qquad \begin{cases} \frac{dz}{ds} = z \\ z_0(s=0) = \tau \end{cases} \quad (10)$$

Note that (8) & (10) coupled, so using elimination of variables for system of ODE's, combine (8) & (10):

$$\begin{cases} \frac{dx}{ds} = z \\ \frac{dz}{ds} = z \\ z_0(s=0) = \tau \\ x_0(s=0) = \tau \end{cases}$$

$$\xRightarrow[\text{of vars}]{\text{sep.n}} \int \frac{dz}{z} = \int ds + \ln C_1 \Rightarrow \ln(z) = \ln(e^s \cdot C_1) \Rightarrow \boxed{z = e^s \cdot C_1} \text{ then}$$

$$\frac{dx}{ds} = e^s \cdot C_1 \Rightarrow \int dx = \int e^s \cdot C_1 \, ds + C_2 \Rightarrow \boxed{x = C_1 e^s + C_2}$$

So, using the ICs:

$$z_0(s=0) = e^0 \cdot C_1 = \tau \Rightarrow \boxed{C_1 = \tau}$$

$$x_0(s=0) = \tau \cdot e^0 + C_2 = \tau \Rightarrow \boxed{C_2 = 0} \quad \text{So}$$

(11) $\boxed{x(s) = \tau e^s}$

(12) $\boxed{z(s) = \tau e^s}$ and finally by (9),

$\therefore \frac{dy}{ds} = 1$ using sep-n of var-s.

$\int dy = \int ds + C_3$

$y = s + C_3$

$y_0(s=0) = 0 + C_3 = 0 \Rightarrow C_3 = 0 \Rightarrow$

(13) $\boxed{y = s}$ ✓ &

(11) $\boxed{x(s) = \tau e^s}$

(12) $\boxed{z(s) = \tau e^s}$ ✓

⑤ Eliminate s, τ in (11), (12) (13):

$\boxed{s = y} \Rightarrow x(y) = \tau e^y \Rightarrow \boxed{\tau = \frac{x}{e^y}}$

$z = \tau e^s = \frac{x}{e^y} \cdot e^y = x$

(13) $\boxed{z = x}$ ✓

⑥ Drawing characteristics $\langle x(s), y(s) \rangle$, No need.

(c) Checking the sol-n in (13)

IC: $z(x,0) = x \equiv x$ ✓

Eq-n: $z \cdot \underbrace{z_x}_{1} + \underbrace{z_y}_{0} = z \cdot 1 + 0 = z \equiv z$ ✓

good!

$\boxed{5_3}$

In summary (running out of time otherwise draw the y(x) graph)

along each charateristic curve (A), (B), (C), (D)

$\tau=-1$ $\tau=0$ $\tau=1$ $\tau=2$

τ is fixed $\Rightarrow -\dfrac{1}{\sin^2(\tau)} = \text{const}$

$$\frac{\tau}{\sin^2(\tau)} = \text{const}$$

charac(.c)

so they are lines:

$\sin(2) > \sin(120°) =$

$\sin(60°) \simeq \frac{\sqrt{3}}{2} = 0.87$

$\sin^2(1) \simeq 0.64$

$1 \simeq 57.3$

$$y_{\tau=-1} = \frac{-1}{\sin^2(-1)}x + \frac{(-1)}{\sin^2(-1)}$$

$$\frac{-1}{\sin^2(1)} = \frac{-1}{0.64} \simeq -1.1$$

$$y_{\tau=0} \Rightarrow x=0$$

(since $x-\tau = a(f(\tau))\cdot y$
$x-0 = a(\sin(0))\cdot y$
$x-0 = -0^2\cdot y$ $\boxed{x=0}$)

? Trig ?

$$y_{\tau=1} = \frac{-1}{\sin^2(1)}x + \frac{1}{\sin^2(1)}$$

$\sin(1) < \sin(2)$

same so ||.

$\sin^2(1) < \sin^2(2)$

$$y_{\tau=2} \quad \frac{-1}{\sin^2(2)}x + \frac{2}{\sin^2(2)}$$

$\dfrac{-1}{\sin^2}$ good!

$\dfrac{1}{\sin^2(1)} > \dfrac{1}{\sin^2(2)}$

$-\dfrac{1}{\sin^2(1)} < -\dfrac{1}{\sin^2(2)}$

showing steeper!

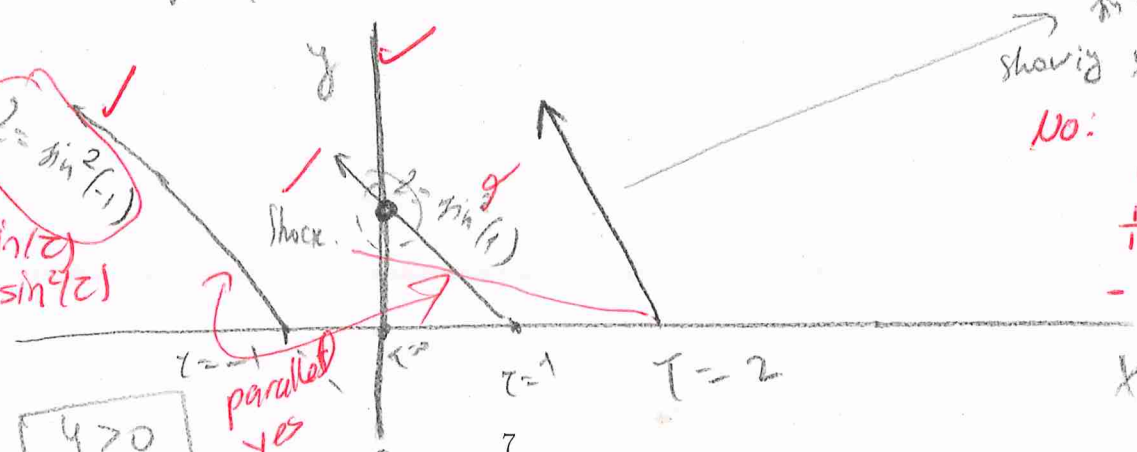
No:

$1 < 2$

$\frac{1}{1} > \frac{1}{2}$

$-1 < -\frac{1}{2}$

less steep



given $\boxed{y>0}$

(6₂)

Based on what I graph in a fine rush there's a shock! that's answers (c)

(c) I could check algebraically for clear & in detail

implicit expression doesn't help much

$\boxed{F_2 > 0}$

answer for (c)! Run out of time

IFT:

Let $F(x,y,z) \in C^1(\mathbb{R}^3)$ (can be redefined). If

$\left.\frac{\partial F}{\partial z}\right|_{(x_0,y_0,z_0)} \neq 0$, then $F(x,y,z)=C$ "can be solved for z"

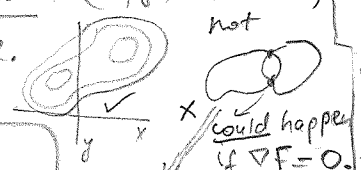
in terms of x, y, z in a nbd (open & connected set containing IC.) of (x_0,y_0,z_0), i.e. $z=f(x,y)$ &

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \& \quad z_y = -\frac{F_y}{F_z}.$$

Level Surface Thm.

Suppose $F(x,y,z) \in C^1(\Omega)$, $\Omega \subseteq \mathbb{R}^3$, & $\nabla F(x,y,z) = \langle F_x, F_y, F_z \rangle$ By(3)

If $\nabla F \neq \langle 0,0,0 \rangle$ for all $(x,y,z) \in \Omega$, then $F(x,y,z)=C$ is a level surface, and each $(x_0,y_0,z_0) \in \Omega$, belongs to only one level surface.



PDE Clarification.

(i) Quasi-linear: $P(x,y,z)z_x + Q(x,y,z)z_y = R(x,y,z)$

(ii) Almost-linear: $P(x,y)z_x + Q(x,y)z_y = R(x,y,z)$

(iii) Linear: $P(x,y)z_x + Q(x,y)z_y + R(x,y)\cdot z = d(x,y)$

(iv) Nonlinear: Not (i), (ii) or (iii).

Leibniz Rule: (Combination of FToC & Chain rule)

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(x,t)\,dt = \int_{g(x)}^{h(x)} \frac{\partial}{\partial x} f(x,t)\,dt + f(x,h(x))\cdot h'(x) - f(x,g(x))\cdot g'(x).$$

Conservation Law:

Eq-n: $a(z)\cdot z_x + z_y = 0$

IC: $z(x,0) = f(x)$

Sol-n: $z = f(x - a(z)y)$ (6)

By (i) above, Identifying P, Q, R: ② $P = a(z)$, $Q=1$, $R=0$ & ① Param IC $\langle \tau, 0, f(\tau) \rangle$

⇒ ③ Check for ex-uniq-s:

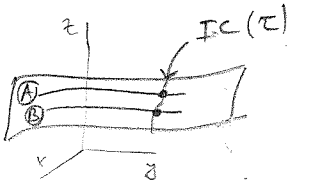
(i) $P, Q, R\ (x_0(\tau), y_0(\tau), z_0(\tau)) \in C^1(\Omega)$. ✓

(ii) $\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0'(\tau) & y_0'(\tau) \end{vmatrix} = -1 \neq 0$. OK ✓

④ Charac-c curves:

$\frac{dx}{ds} = P$, $x(s=0)=x(\tau)$; $\frac{dy}{ds} = Q$, $y(s=0)=y(\tau)$; $\frac{dz}{ds} = R$, $z(s=0)=z_0(\tau)$ ⇒ $\begin{cases} \frac{dx}{ds} = a(z) \\ x_0(s=0) = \tau \end{cases}$, $\begin{cases} \frac{dy}{ds} = 1 \\ y_0(s=0) = 0 \end{cases}$, $\begin{cases} \frac{dz}{ds} = 0 \\ z_0(s=0) = f(\tau) \end{cases}$ ⇒

⇒ $x = a(f(\tau))\cdot s + \tau$ (1) ; $y = s$ (2) ; $z = f(\tau)$ (3)



⑤ Elim-e s, τ in (1), (2), (3) so that $z = f(x,y)$:

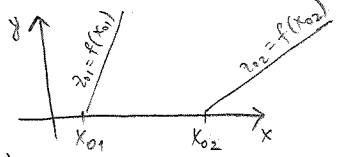
By (2), $s = y \xrightarrow[(1)]{sub} x = a(f(\tau))\cdot y + \tau$,

Along each charac-c curve Ⓐ, Ⓑ, ..., τ is fixed. $z = f(\tau) = z_0$, $\tau = const. = x_0$.

⇒ $x = a(z_0)\cdot y + x_0$ ⇒ $\boxed{x - x_0 = a(z_0)\cdot y}$ (4) = Characteristics eq-n on xy plane.

⑥ Drawing characteristics $(x(s), y(s))$ using (4): $\boxed{y = \frac{1}{a(z_0)}\cdot(x - x_0)}$ (5)

(i) $a(z)$ is ↓

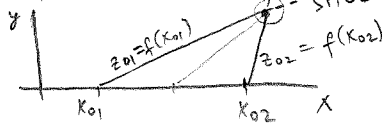


Say $f(x)$ is ↑, $z = f(\tau) = f(x_0)$ ⇒ $z_{01} < z_{02}$

⇒ $a(z_{01}) < a(z_{02})$

$\frac{1}{a(z_{01})} > \frac{1}{a(z_{02})}$

(ii) $a(z)$ is ↓ (Say $f(x)$ is still ↑) - SHOCK!



$z_0 = f(\tau) = f(x_0)$ ⇒ $z_{01} < z_{02}$

⇒ $a(z_{01}) > a(z_{02})$

$\frac{1}{a(z_{01})} < \frac{1}{a(z_{02})}$

Get a shock unless $z_{01} = z_{02}$ or $f(x_{01}) = f(x_{02})$.

⑦ Check sol-n

Eq-n: $a(z)\cdot z_x + z_y = 0$ ✓ ; IC: $z(x,0) = f(x)$ ✓

* Algebraic way of determining SHOCK:

Apply F_z on sol-n $z - f(x - a(z)y) = 0$ (6) for IFT:

$F_z = 1 + f'(x - a(z)y)\cdot a'(z)y > 0$.

Conservation law: $\frac{\partial d}{\partial t} + c[1-2d]\frac{\partial d}{\partial x} = 0$ with IC: $d(x,0) = f(x)$.

Sol-n: $d = f(x - ct(1-2d)) \xrightarrow[\text{sol-n exist}]{\text{IFT}} F_d = 1 - 2ct f'(x - ct(1-2d)) > 0$

Characteristics: $d = d_0$ on $\boxed{x - x_0 = c[1-2d_0]t}$ (using (4) above), $x_0 = \tau$ as always for Cons. Law

Build $ct = ct(x)$ rel-n.

graphing sol-ns (?) $d = 0$ when $ct = 0$ & $x \geq \frac{1}{f'}$...?

Lazizbek S